

DIVYANSHU MATHUR

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Education

Vellore Institute of Technology (VIT)

Bachelor of Technology in *Computer Science and Engineering* - 9.14 CGPA

2022 – 2026

Vellore, Tamil Nadu

Lucky Bal Niketan Senior Secondary School

Class XII (CBSE) - 92.6%

2021 – 2022

Jodhpur, Rajasthan, India

Lucky Bal Niketan Senior Secondary School

Class X (CBSE) - 90.4%

2019 – 2020

Jodhpur, Rajasthan, India

Technical Skills

Programming Languages: Python, R, C++, C, Java, JavaScript, HTML, CSS

Libraries & Frameworks: NumPy, Pandas, Scikit-learn, TensorFlow, Keras, PyTorch, NLTK, Matplotlib, Seaborn, Streamlit, React.js, Flask, OpenCV, LangChain, LangGraph

Tools & Platforms: Selenium, Power BI, Tableau, Excel, Git, GitHub

Databases: MySQL, PostgreSQL, SQL Plus

Core Concepts: Data Structures & Algorithms, Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Exploratory Data Analysis, Retrieval-Augmented Generation

Experience

Indian Institute of Technology (IIT) Jodhpur

Research Intern (Data Analyst)

May 2025 – Jul 2025

Jodhpur, Rajasthan

- Analyzed urban mobility dataset containing 50,000+ records to identify commuter behavior patterns and transportation preferences.
- Engineered behavioral and demographic features from high-dimensional survey data, improving model readiness
- Conducted exploratory data analysis and statistical testing to uncover significant demographic predictors influencing travel decisions.
- Applied clustering techniques to identify 5 commuter personas enabling behavioral segmentation analysis.
- Built supervised machine learning models to predict transport mode achieving 87% accuracy

Aamby Valley City Limited

Web Development Intern

Jun 2024 – Jul 2024

Pune, Maharashtra

- Developed responsive, mobile-first web interfaces, improving load time and performance by 30%.
- Ensured cross-browser and cross-device compatibility while translating UI/UX wireframes into interactive web pages.

Projects

Real-Time Vehicle Speed Estimation & License Plate Recognition | [GitHub](#)

- Designed a real-time traffic monitoring system using YOLOv11 for vehicle detection and DeepSORT for robust multi-object tracking
- Implemented geometric calibration and temporal smoothing reducing speed estimation noise
- Integrated OCR-based license plate recognition using EasyOCR achieving 71% recognition accuracy.
- Designed modular PyTorch and OpenCV pipeline supporting real-time video analytics and export functionality.

Sentiment Analyzer (Speech + Text Emotion Detection) | [GitHub](#)

- Built multimodal emotion classification model using Wav2Vec2 audio embeddings and Sentence-BERT text features on 3,000+ voice samples.
- Achieved 75% Accuracy on emotion classification task using combined feature embeddings.
- Deployed real-time prediction interface via Streamlit with audio file upload and interactive visualization.

Conversational AI Agent | [GitHub](#)

- Developed a stateful Conversational AI Agent using LangGraph, LangChain, Streamlit, and Gemini API.
- Implemented persistent multi-turn chat memory with SQLite and thread-based conversation management.
- Integrated external AI tools including Web Search, Calculator, and Real-Time Stock Price APIs using tool-calling.
- Built a responsive Streamlit chat interface with streaming responses and graph-based workflow orchestration.